

D4 w1 Strict Bound -- NEGATIVE Certificate (2026-06-01)

STATUS: VERIFIED_OUT_OF_TOWER_NEGATIVE

D4 FAILS for beta = 0.86. Requires beta > 2.07942.

DISCLAIMER: Numerical only. NOT Arb/MPFI. NOT Lean. NOT trio-clean. NOT mass gap.

1. Statement under test

$w1(\beta) = \int \{SU(3)\} \exp(-\beta * S(U)) d \text{ haar}(U) < 1/7$, at submitted $\beta = 0.86$.

Repo action: $S(U) = \text{plaquetteEnergy} = (3 - \text{Re tr } U)/3$ (WilsonPositivity.lean:27).

2. Method

Deterministic Weyl integration over the $SU(3)$ maximal torus

$|\Delta|^2 = \prod_{j < k} (2 - 2 \cos(t_j - t_k))$, $t_1, t_2, t_3 = -t_1 - t_2$, self-normalized.

Grid $n = 2400$. Cross-checked vs $2e6$ -draw Haar- $SU(3)$ Monte Carlo.

Validation: $w1(0) = 1.000000$; $w1(0.86) = 0.432367$ (quadrature) vs 0.4324 (MC) -- agree.

3. Result

beta	w1(beta)	< 1/7 ?
0.86	0.432367	NO
1.00	0.378821	NO
1.50	0.238739	NO
2.00	0.153058	NO
2.07942	0.142857 (=1/7)	boundary
2.50	0.099908	YES
3.00	0.066441	YES

4. Findings

1. D4 FAILS at $\beta = 0.86$: $w1(0.86) \approx 0.4324 \approx 3.0 \times (1/7)$.
2. Minimal threshold $\beta_0 \approx 2.07942$ -- bound holds only for $\beta > 2.07942$.
3. Submitted 0.85 was the un-normalized action ($3 - \text{Re tr } U = 3*S$); the repo's $/3$ -normalization scales the threshold by ~ 3 . Submitted value 0.0540459 is reproduced by no tested normalization (withdrawn).

5. Honesty locks / scope

- status_of_this_leaf: VERIFIED_OUT_OF_TOWER_NEGATIVE ; closes_D4: false ; closes_mass_gap: false ; discharges_surface: false ; surface_1_status: OPEN ; ym_tower_status: OPEN.
- NEGATIVE verdict is numerically robust (0.4324 vs 0.1429). β_0 is a numerical estimate, NOT a rigorous Arb/MPFI interval certificate.
- OUT-OF-TOWER / Not Lean / Not trio-clean.
- Even if re-established for $\beta > \beta_0$: supplies only Wall256 hw1; nothing else.
- Does NOT impact MassGap574.lean, Clay YM, or Surface #1 -- no mass-gap / $\mu > 0$ / $m > 0$.
- Does NOT close the Wall256 conditional (hOS, h_bridge OPEN) or KP.

6. Termination

closes_leaf_D4: false. d4_verdict: TESTED NEGATIVE -- submitted bound false at $\beta = 0.86$ under repo action. minimal_beta0_repo_action: 2.07942.

Parent KP surface, Surface #1, and the YM tower stay OPEN; no mass-gap / $m > 0$ / Clay claim.